

Parametric Heatsink Conjugate Heat Transfer Simulation Report

OpenFOAM 13 · foamMultiRun (Buoyant, Multi-Region) · 3-Point Fin-Pitch Sweep
Parallel/Serial Verified · 16,000 Steady-State Iterations per Case

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Parameter	Value
Solver	foamMultiRun (buoyant, 3-region CHT)
Regions	fluid (air); solid → heatsink (Al) + cpu (Si) cell zones
CPU heat dissipation	150 W
Fan	Delta FFB0812VH (quadratic curve, fanPressure BC)
Fan outlet velocity (converged)	3.87 m/s
Inlet air temperature	300 K (27°C)
Verification	Serial vs. 8-core parallel agree to 0.001 K (5 mm baseline)
Sweep	3 mm (24 fins) / 4 mm (18 fins) / 5 mm (15 fins)
Best CPU temperature	336.96 K (63.96°C) -3 mm pitch, 24 fins
Worst CPU temperature	343.90 K (70.90°C) -5 mm pitch, 15 fins
Temperature reduction (5→3 mm)	6.94 K

1. Project Overview

This project presents a steady-state Conjugate Heat Transfer (CHT) parametric fin-pitch sweep for a CPU and fin-array heatsink inside a fan-driven duct, using OpenFOAM 13's foamMultiRun multi-region driver. Unlike a single fixed-geometry case, the mesh itself is generated parametrically: a Python script lays out the fin/channel geometry directly as a blockMeshDict from a fin pitch and fin count, so the design can be swept without rebuilding CAD geometry by hand. Three cases were run -5 mm, 4 mm, and 3 mm fin pitch -to identify the thermal optimum for this fan/duct combination under 150 W CPU dissipation.

Key Features

- 3-region CHT: fluid air domain, aluminium heatsink, and silicon CPU die, coupled via coupledTemperature interfaces
- Fully parametric blockMeshDict generation (gen_cases_v2.py) -fin pitch and fin count are input parameters, not hand-edited geometry
- Fan modelled with a fanPressure boundary condition driven by a tabulated quadratic fan curve (Delta FFB0812VH), not a fixed-velocity inlet
- Volumetric heatSource fvModel applies 150 W directly to the CPU silicon cell zone
- Baseline 5 mm case cross-checked on 1 core (serial) and 8 MPI ranks -converged results agree to 0.001 K
- Three-point pitch sweep completed: 5 mm / 4 mm / 3 mm, each run to 16,000 steady-state iterations
- Diminishing-returns trend identified: 5→4 mm saves 5.84 K; 4→3 mm saves an additional 1.10 K

2. Geometry and Mesh

2.1 Parametric Geometry Generation

The heatsink/duct geometry is produced directly as an OpenFOAM blockMeshDict by gen_cases_v2.py, rather than imported from CAD. Given a fin pitch and fin count, the script lays out alternating fin / channel / side-gap strips across the 80 × 80 mm duct cross-section, splits those strips against the CPU footprint, and writes the resulting hex-block mesh with fan, wall, and region-zone tagging already built in. Each pitch case is a fresh blockMesh run from the same generator with different parameters -no manual geometry editing between cases.

Fixed geometry: duct cross-section 80 × 80 mm; fin thickness 1.5 mm; fin height 35 mm; base thickness 5 mm; CPU die 40 × 40 × 4.5 mm centred under the fin array; inlet and outlet plenum length 105 mm each (total duct length 290 mm). Side gaps are sized to fill the remaining duct width symmetrically for each pitch.

2.2 Per-Case Geometry

Case	Fin Pitch	Fins	Channel Width	Approx. Mesh Cells
p5_cpu	5 mm	15	3.5 mm	649,200
p4_cpu	4 mm	18	2.5 mm	~695,520
p3_cpu	3 mm	24	1.5 mm	~780,000

2.3 Mesh

Each case uses a block-structured hex mesh built with blockMesh and split into coupled regions using splitMeshRegions -cellZones. After splitting, topoSet -region cpu is run to recreate the allCpuCells cellZone

(required by the heatSource fvModel) which is not preserved across a region split.

Region	Material	Role
fluid	Air (buoyant, k-ε)	Flow & heat transport
heatsink	Aluminium (k = 205 W/m·K, ρ = 2700 kg/m³)	Heat spreading / dissipation
cpu	Silicon (k = 150 W/m·K, ρ = 2329 kg/m³)	Heat source (150 W via heatSource fvModel)

3. Simulation Setup

3.1 Solver Configuration

Each case is driven by foamMultiRun, coupling one buoyant fluid region (solved with k-ε turbulence and a p_rgh pressure formulation) to a solid super-region containing the heatsink and cpu cell zones. Region interfaces use the coupledTemperature boundary condition at every fluid↔heatsink, fluid↔cpu, and heatsink↔cpu face. Heat is injected directly into the CPU silicon zone using a heatSource fvModel (cellZone allCpuCells, Q = 150 W).

Relaxation factors are set to p_rgh 0.4, U 0.6, h/k/ε 0.4 across all three cases -a conservative choice validated after the finer-pitch meshes were found to diverge at higher values (0.7–0.8). End time is 16,000 iterations, reduced from 20,000 after the 5 mm baseline confirmed convergence plateau by ~15,000 steps.

3.2 Fan Boundary Condition

Rather than a fixed-velocity inlet, the fan is modelled with a fanPressure boundary condition at the fan1_half0 patch, driven by a tabulated pressureVsQ.csv curve. The table is generated from the quadratic Delta FFB0812VH fan curve $dP = 73.99 - 1164.25 \cdot Q - 34615 \cdot Q^2$ [Pa, Q in m³/s], so the operating point of the fan against each duct's actual flow resistance is solved for -not assumed. As fin pitch tightens, flow resistance increases and the operating point shifts to lower flow rates.

3.3 Boundary Conditions Summary

Boundary	Field	Type	Value
fan1_half0 (inlet)	p_rgh	fanPressure	Delta FFB0812VH curve (table)
fan1_half0 / fan2_half0	U	pressureInletOutletVelocity	Solved from fan curve
base_bottom / top_wall / side_walls	U	noSlip	-
fluid↔heatsink, fluid↔cpu, heatsink↔cpu	T	coupledTemperature	Region-coupled
CPU cell zone	heatSource (fvModel)	heatSource	150 W

3.4 Verification: Serial vs. Parallel (5 mm Baseline)

The 5 mm-pitch baseline case was run twice to 20,000 iterations -once on a single core and once decomposed across 8 MPI ranks -to confirm that domain decomposition does not change the converged solution. Both runs converge to T_max = 343.900 K, agreeing to within 0.001 K.

4. Results

4.1 Parametric Sweep Summary

All three pitch cases ran to 16,000 steady-state iterations and reached convergence well before end time. CPU maximum temperature was monitored every iteration via the `cpuTmax volFieldValue` function object. The results show a clear monotonic decrease in CPU temperature as fin pitch decreases, with strongly diminishing returns below 4 mm.

Case	Fin Pitch	Fins	T_max (K)	T_max (°C)	ΔT vs. ambient	ΔT vs. 5 mm
p5_cpu	5 mm	15	343.90	70.90	43.90 K	- (baseline)
p4_cpu	4 mm	18	338.06	64.06	38.06 K	−5.84 K
p3_cpu	3 mm	24	336.96	63.96	36.96 K	−6.94 K

4.2 Convergence

Each case's `T_max` history (available in `postProcessing/cpu/cpuTmax/*/volFieldValue.dat`) shows a smooth monotonic climb from the initial condition and a clear plateau well before end time. The 5 mm baseline plateaus at ~15,000 iterations; the finer-pitch cases (larger meshes) follow the same pattern within the same 16,000-iteration budget. Residuals for `U`, `h`, `e` (both solid regions), and `p_rgh` are all below 1×10^{-6} at convergence.

4.3 Physical Interpretation

The dominant effect of tighter fin pitch is increased total fin surface area (15 → 18 → 24 fins), which enhances convective heat transfer and reduces CPU temperature. The opposing effect is increased channel flow resistance, which shifts the fan operating point to lower flow rates. Between 5 mm and 4 mm, the surface area gain dominates strongly (−5.84 K). Between 4 mm and 3 mm, the channel width narrows to 1.5 mm and the resistance penalty begins to erode the surface area benefit, yielding only −1.10 K. This strongly diminishing return indicates the thermal optimum for this fan/duct combination lies in the 3–4 mm range.

A further tightening to 2 mm (channel width 0.5 mm) was considered but not pursued: at that geometry the fan would operate at significantly reduced flow, and the net thermal benefit is expected to be negligible or negative -the 3 mm case already shows the trend turning.

4.4 Workflow

For each pitch case: `gen_cases_v2.py` (parametric `blockMeshDict`) → `blockMesh` (hex mesh with fin/channel/side-gap strips and fan/wall patches) → `splitMeshRegions -cellZones` (fluid | heatsink | cpu) → `topoSet -region cpu` (recreate allCpuCells cellZone) → `foamMultiRun` (coupled buoyant + conductive solve, 16,000 iterations) → `postProcessing` (cpuTmax, fan patch flow rate and velocity).

5. Status

All three cases of the fin-pitch sweep are complete:

Case	Fin Pitch	Fins	Status	T_max
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p5_cpu (serial)	5 mm	15	Converged, 20,000 it.	343.90 K
p5_cpu (8-core parallel)	5 mm	15	Converged, 20,000 it.	343.90 K
p4_cpu	4 mm	18	Converged, 16,000 it.	338.06 K
p3_cpu	3 mm	24	Converged, 16,000 it.	336.96 K

6. Limitations

- Region coupling uses a simplified coupledTemperature boundary condition rather than a baffle-resistance formulation, so thermal interface material (TIM) contact resistance between the CPU die and heatsink base is not explicitly represented.
- Outer duct boundaries (base_bottom, top_wall, side_walls) use noSlip walls rather than a true open-atmosphere condition; the duct cross-section fully constrains the flow domain.
- The fan curve table is a quadratic fit to the published Delta FFB0812VH specification curve rather than measured operating data for this specific duct impedance.
- All three cases were run serially (single core) to avoid resource conflicts; wall-clock times of ~22 hours per case are acceptable for a portfolio study but would benefit from parallelisation in a production setting.
- The sweep covers three pitch points; the resolution is sufficient to identify the trend and the approximate optimal range (3–4 mm) but not to pinpoint a precise optimum to sub-millimetre accuracy.

7. Methodology Summary

- **Geometry:** Parametric blockMeshDict generation via gen_cases_v2.py (Python) -fin pitch and fin count as direct inputs; one fresh mesh per pitch case.
- **Mesh:** blockMesh hex mesh with fin/channel/side-gap strip decomposition; splitMeshRegions -cellZones for fluid/heatsink/cpu region split; topoSet -region cpu to recreate allCpuCells cellZone after each split.
- **Region Coupling:** coupledTemperature boundary condition at all fluid↔solid and solid↔solid interfaces.
- **Heat Source:** heatSource fvModel, cellZone allCpuCells, $Q = 150$ W.
- **Fan Modelling:** fanPressure boundary condition with tabulated quadratic Delta FFB0812VH curve (pressureVsQ.csv).
- **Solver:** foamMultiRun (buoyant fluid + conductive solid, k - ϵ turbulence), 16,000 steady-state iterations; relaxation p_rgh 0.4 / U 0.6 / h-k- ϵ 0.4.
- **Verification:** Baseline 5 mm case run serial and on 8 MPI ranks; converged T_max agrees to 0.001 K.
- **Post-processing:** cpuTmax volFieldValue function object; fan patch flow rate and velocity monitors.

Software Stack

Tool	Version	Purpose
OpenFOAM	13	CFD solver (foamMultiRun, buoyant multi-region CHT)
Python	3.x	Parametric blockMeshDict generation (gen_cases_v2.py)
Ubuntu (WSL2)	24.04	Operating system
OpenMPI	-	8-core parallel verification run (5 mm baseline)
reportlab	-	PDF report generation

8. Conclusion

A three-point parametric fin-pitch sweep (5 mm / 4 mm / 3 mm) was completed using OpenFOAM 13's foamMultiRun multi-region CHT driver with a fully parametric mesh generation pipeline. All three cases dissipate 150 W from a CPU die into a fan-driven duct with the Delta FFB0812VH fan modelled via its published pressure curve rather than a fixed flow rate.

The sweep demonstrates a clear and physically intuitive result: tightening fin pitch from 5 mm to 4 mm yields a strong 5.84 K reduction in peak CPU temperature (343.90 K $\rightarrow$ 338.06 K) driven by increased fin surface area. A further tightening to 3 mm adds only 1.10 K of improvement (338.06 K $\rightarrow$ 336.96 K), indicating that channel flow resistance is beginning to offset the surface area benefit. The thermal optimum for this fan/duct combination lies in the 3–4 mm range; further pitch reduction is expected to yield negligible or negative returns.

The full case files, including the parametric mesh generator and solver configuration, are available at github.com/samuelsimmons99/OpenFOAM-Portfolio.